Automated Student Attendance Monitoring & Analytics System For Colleges

A smart, data-driven solution for colleges

Presented By

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Subject: DTIE-Projects

01 Problem

- Attendance marking during class is time-consuming.
- Paper registers cause errors and allow proxy attendance.
- Faculty calculate attendance percentages by hand.
- Students and parents learn about shortages too late.
- Students with low attendance are not identified on time.



02 Empathy

Students
- Fear unfair marking
- Want to see their own attendance
- Dread shortage detention

Faculty
- Lose teaching time
- Cannot catch proxies in large classes
- Repeat paperwork

HOD / Admin
- Need timely, accurate reports for eligibility and accreditation

03 Define

How might we record and analyse attendance accurately, with minimal effort, and give every stakeholder timely insight?

Users: Students, Faculty, HODs, Administrators

Needs: Fast capture, proxy prevention, live percentages, alerts

Goal: Attendance in under 1 minute; flag at-risk students early

04 Idea Generation

Dynamic QR · Face recognition · RFID / NFC cards · Biometric · GPS geofencing · Auto SMS alerts

05 Multi Voting

Winner: Dynamic QR + mobile app

(Replace with your team's real votes.)

Votes per idea (illustrative)

Idea	Votes
Dynamic QR + app	14
Face recognition	9
RFID / NFC	6
Biometric	4
GPS geofencing	3

06 Rapid Estimation & Solution Concept

Time to working prototype
6–8 wks

Cost
Low (open-source + free-tier cloud)

Modules
Login, QR, logs, analytics

Concept
Dynamic-QR attendance app + cloud analytics dashboard

Tech Stack
React / Flutter (front end), Node.js + Database (back end)

Risk
Proxy scans & poor network → time-limited codes & offline sync

07 Key Features & Analytics

- ✓ Live attendance percentage per student, subject and class
- ✓ Automatic alerts when attendance falls below 75%
- ✓ Trend charts and defaulter lists for faculty and HODs
- ✓ One-click reports for exams, parents and accreditation
- ✓ Secure role-based access for students, faculty and admin

08 Rough Prototype

Faculty starts session → QR code shown → Students scan → Saved to database → Dashboard & alerts



09 System Architecture / Working Flow

Student Identification → Attendance Capture → Attendance Verification → Secure Database → Attendance Processing → Analytics Dashboard → Reports & Alerts

Users: Admin · Faculty · Students

10 Attendance Analytics

Turning Attendance Data into Useful Insights

Individual Attendance
Attendance Percentage = Classes Attended ÷ Classes Conducted × 100

Subject-Wise Analysis
Compare attendance across different subjects.

Trend Analysis
Track weekly or monthly attendance patterns.

Attendance Status	% Range
Good	85% and above
Monitor	75% – 84%
Warning	Below 75%

11 Demonstration

1. Faculty logs in and opens a class session
2. A time-limited QR code is generated
3. Student scans it and is marked present instantly
4. Dashboard shows live percentage and flags < 75% students
5. Report and alert sent to HOD and parents

12 Conclusion & Future Scope

- ✓ Saves class time and removes manual errors
- ✓ Prevents proxy attendance with rotating QR codes
- ✓ Gives real-time analytics for early intervention

Future: Face recognition, LMS integration, predictive dropout alerts

Thank You